

Reviewer Pack — Minimal Frobenius-Schur Indicator and Communication Complexi...

Entry fad0317c072b · INCONCLUSIVE

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Minimal Frobenius-Schur Indicator and Communication Complexity Rank Variance

Entry ID: fad0317c072b **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 22:03:00 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory **Field B** (complexity object): Communication Complexity

Statement:

For every CNF φ with m clauses over n variables, the variance of the Frobenius-Schur indicator over all irreducible representations V of the symmetric group S_n is lower-bounded by the square of the communication complexity rank ($\text{ccr}(\varphi)$) of φ , i.e., $\text{Var}(S_n, \varphi) \geq c^2 * \text{ccr}(\varphi)$, where c is a constant.

Rationale (proposer's reasoning):

The Frobenius-Schur indicator captures the non-commutativity in group representations, which can be related to the complexity of distributing information in communication complexity. The variance of this indicator may reflect the structural complexity of the problem instance, as measured by the communication complexity rank.

Taxonomy category: REPRESENTATION_THEORY_X_COMMUNICATION_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b3490abb4c3461e1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given CNF φ with m clauses over n variables, $\text{Var}(S_n, \varphi)$ is considered supported if $\text{Var}(S_n, \varphi) \geq c^2 * \text{ccr}(\varphi)$ AND $\text{support_fraction} \geq 0.8$ WHERE support_fraction = the fraction of seeds that meet this condition.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Frobenius-Schur indicator" AND "communication complexity rank variance" AND representation theory - symmetric group S_n AND irreducible representations AND communication complexity - "Frobenius-Schur indicator" variance $\geq c^2 * ccr(\phi)$ AND CNF ϕ

Top relevant hits considered: - [http://arxiv.org/abs/hep-ph/0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [http://arxiv.org/abs/0902.2349v1] Defect of characters of the symmetric group - [http://arxiv.org/abs/1908.06409v1] Schur multipliers of special p-groups of rank 2

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n, m):
        clauses = []
        for _ in range(m):
            clause = [random.randint(1, n), -random.randint(1, n)]
            clauses.append(clause)
        return clauses

    def frobenius_schur_indicator(cnf):
        # Placeholder implementation
        return 0.5

    def communication_complexity_rank(cnf):
        # Placeholder implementation
        return len(cnf)

    def variance(values):
        mean = sum(values) / len(values)
        return sum((x - mean) ** 2 for x in values) / len(values)

    n = random.randint(5, 40)
    m = random.randint(n, 3 * n)
    cnf = generate_cnf(n, m)

    indicator = frobenius_schur_indicator(cnf)
```

```

    ccr = communication_complexity_rank(cnf)
    var = variance([indicator] * len(indicators))

    return {
        "metric_name": "Var(Sn,  $\phi$ )",
        "metric_value": var,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
        31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
        73, 79, 83, 89, 97, 101, 103, 107, 109, 113
    ]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_var = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_var} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_var} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d4d6c36.py", line 66, in <module>

```

    result = run_trial(seed)
    ~~~~~
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d4d6c36.py", line 46, in run_trial

```

    var = variance([indicator] * len(indicators))
    ~~~~~
```

NameError: name 'indicators' is not defined. Did you mean: 'indicator'?

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed due to a `NameError`, which prevented it from producing any data or results. | next: Review the test code for the definition of ‘indicators’ and ensure that all necessary variables are defined before use.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14339
2	preregistration	ollama_remote	glm4:latest	0	0	9603
3	novelty	ollama_remote	glm4:latest	0	0	8685
4	novelty	ollama_remote	glm4:latest	0	0	9691
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13750
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8527
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6951
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7699
9	judge	ollama_remote	glm4:latest	0	0	17161

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 96408 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/fad0317c072b.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/fad0317c072b.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/fad0317c072b.tar.gz` (if generated)